# Online Supplementary Material

"*On Synthetic Difference-in-Differences and Related Estimation Methods in Stata*"

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## A1 Estimation Algorithms for the Block Design

In this Appendix, we replicate the estimation algorithm and inference algorithms defined in Arkhangelsky et al. (2021). These are referred to in the text, and follow the same notation as in Section 2 of the paper.

**Algorithm A1:** Algorithm 1 from Arkhangelsky et al. (2021)

Data: $\mathbf{Y}$, $\mathbf{W}$.
Result: Point estimate $\widehat{\tau}^{sdid}$.

1. Compute regularization parameter $\zeta$;
2. Compute unit weights $\hat{\omega}^{sdid}$;
3. Compute time weights $\hat{\lambda}^{sdid}$;
4. Compute the SDID estimator via the weighted DID regression;

$$\left(\hat{\tau}^{sdid}, \hat{\mu}, \hat{\alpha}, \hat{\beta}\right) = \underset{\tau,\mu,\alpha,\beta}{\arg\min} \left\{ \sum_{i=1}^{N} \sum_{t=1}^{T} (Y_{it} - \mu - \alpha_i - \beta_t - W_{it}\tau)^2 \widehat{\omega}_i^{sdid} \hat{\lambda}_t^{sdid} \right\}$$

**Algorithm A2:** Algorithm 2 from Arkhangelsky et al. (2021)

Data: $\mathbf{Y}$, $\mathbf{W}$, $B$.
Result: Variance estimator $\widehat{V}_{\tau}^{cb}$

**for** $b \leftarrow$ *1 to* $B$ **do**
  1. Construct a bootstrap dataset $(\mathbf{Y}^{(b)}, \mathbf{W}^{(b)})$ by sampling $N$ rows of $(\mathbf{Y}, \mathbf{W})$ with replacement;
  2. **if** *the bootstrap sample has no treated units or no control units* **then**
     Discard resample and go to 1;
     **end**
  3. Compute SDID estimate $\tau^{(b)}$ following algorithm A1 based on $(\mathbf{Y}^{(b)}, \mathbf{W}^{(b)})$;

**end**

4. Define $\widehat{V}_{\tau}^{cb} = \frac{1}{B} \sum_{b=1}^{B} \left( \widehat{\tau^{(b)}} - \frac{1}{B} \sum_{b=1}^{B} \widehat{\tau^{(b)}} \right)^2$;

**Algorithm A3:** Algorithm 3 from Arkhangelsky et al. (2021)

Data: $\widehat{\omega}$, $\widehat{\lambda}$ $\mathbf{Y}$, $\mathbf{W}$, $\widehat{\tau}$.
Result: Variance estimator $\widehat{V}_\tau$

**for** $i \leftarrow$ *1 to* $N$ **do**
1. Compute $\widehat{\tau}^{(-i)}$ : $\arg\min_{\tau,\{\alpha_j,\beta_t\}_{j\neq i,t}} \sum_{j\neq i,t}(Y_{it} - \alpha_j - \beta_t - W_{it}\tau)^2 \widehat{\omega}_j \hat{\lambda}_t$;

**end**

2. Compute $\widehat{V}_\tau^{jack} = (N-1)N^{-1}\sum_{i=1}^{N}\left(\widehat{\tau}^{(-i)} - \widehat{\tau}\right)^2$;

**Algorithm A4:** Algorithm 4 from Arkhangelsky et al. (2021)

Data: $\mathbf{Y}_{co}$, $N_{tr}$, $B$.
Result: Variance estimator $\widehat{V}_\tau^{placebo}$

**for** $b \leftarrow$ *1 to* $B$ **do**
1. Sample $N_{tr}$ out of the $N_{co}$ control units without replacement to 'receive the placebo';
2. Construct a placebo treatment matrix $\mathbf{W}_{co}^{(b)}$, for the controls;
3. Compute SDID estimate $\tau^{(b)}$ based on $(\mathbf{Y}_{co}, \mathbf{W}_{co}^{(b)})$;

**end**

4. Define $\widehat{V}_\tau^{placebo} = \frac{1}{B}\sum_{b=1}^{B}\left(\widehat{\tau^{(b)}} - \frac{1}{B}\sum_{b=1}^{B}\widehat{\tau^{(b)}}\right)^2$;

## A2 Replicating Weight Graphs

After implementing the `sdid` estimator, the unit specific weights can be used to re-create the weight graph provided as output automatically with the `graph` option. While this is somewhat involved, and likely would not be conducted by hand, it may be illustrative to see how this is generated, combining both unit-specific weights, and unit-specific difference-in-difference estimates. This code is displayed below, first saving time weights which are used to calculate DID estimates, secondly saving unit weights for graphing, thirdly combining all elements and calculating the DID estimates, and finally, generating the graph, which is displayed after this code excerpt.

```
. bys state: egen y1=mean(packspercapita) if year>=1989
. bys state: egen y2=mean(packspercapita*lambda) if year<=1988
. replace y2=y2*19
. egen ypost=mean(y1), by(state)
. egen ypre=mean(y2), by(state)
. keep state ypre ypost omega
. duplicates drop
. gen delta=ypost-ypre
. qui sum delta if state=="California"
. gen sdelta=`r(mean)´
. gen difference=sdelta-delta
. egen id=group(state)
. replace omega=. if state=="California"
. drop if omega==.
. encode state, gen(state2) l(id)

. tw scatter difference state2 if omega!=0 [aw=omega], msize(tiny) ||
>     scatter difference state2 if omega==0, m(X)
>     xlabel(1(1)38, angle(vertical) labs(vsmall) valuelabel)
>     yline(-15.60383, lc(red)) legend(off) xtitle("")
>    ytitle("Difference") scheme(sj)
. graph export sdid_panela.eps, replace
```

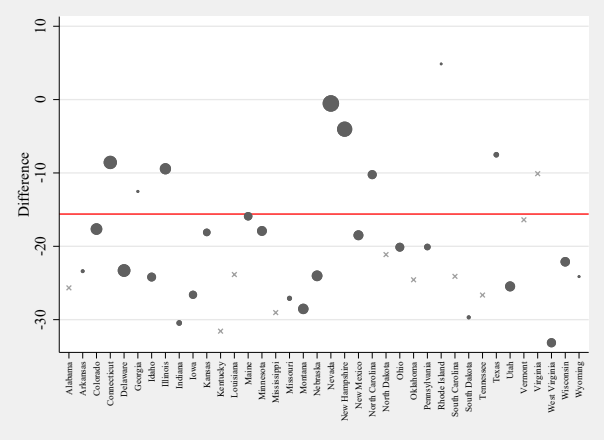



Figure A1: Unit-Specific Weight Graph

## A3 Computational Implementation of the SDID Estimator

While $\widehat{\tau}^{sdid}$ in (1) from the paper is presented as the solution to a two-way fixed effects regression, in practice this parameter can be estimated identically – though in a considerably more computationally efficient manner – directly as a weighted double-differencing estimator. To see this, note:

$$\widehat{\tau}^{sdid} = \left(\bar{Y}^{sdid}_{tr,post} - \bar{Y}^{sdid}_{tr,pre}\right) - \left(\bar{Y}^{sdid}_{co,post} - \bar{Y}^{sdid}_{co,pre}\right)$$

Here $\bar{Y}^{sdid}$ refers to weighted averages generated with SDID weights, *tr* refers to treatment, *co* to control, *pre* to the pre-treatment period, and *post* to the post-treatment period. Note (as in Arkhangelsky et al. (2021, eq. 16)), that this can be re-written as:

$$\widehat{\tau}^{sdid}(\omega, \lambda) = \omega'_{tr}\mathbf{Y}_{tr,post}\lambda_{post} - \omega'_{co}\mathbf{Y}_{co,post}\lambda_{post} - \omega'_{tr}\mathbf{Y}_{tr,pre}\lambda_{pre} + \omega'_{co}\mathbf{Y}_{co,pre}\lambda_{pre} \quad \text{(A1)}$$

where $\mathbf{Y}_{tr,post}$ refers to an $N_{tr} \times N_{post}$ matrix, $\mathbf{Y}_{co,post}$ refers to an $N_{co} \times N_{post}$ matrix, and similarly for *pre* periods. Weights $\omega_{tr}$, $\omega_{co}$ and $\lambda_{post}$, $\lambda_{pre}$ are all column vectors of dimension $N_{tr}$, $N_{co}$, $N_{post}$ and $N_{pre}$ respectively.

We denote as $\mathbf{Y}$ the full set of outcomes written in wide form with a single line for each unit and a single column for each time period, ordered such that:

$$\mathbf{Y} = \begin{pmatrix} \mathbf{Y}_{co,pre} & \mathbf{Y}_{co,post} \\ \mathbf{Y}_{tr,pre} & \mathbf{Y}_{tr,post} \end{pmatrix}.$$

By simple algebra, note that (A1) can be written as:

$$\widehat{\tau}^{sdid}(\omega, \lambda) = \begin{pmatrix} -\omega'_{co} & \omega_{tr} \end{pmatrix} \mathbf{Y} \begin{pmatrix} -\lambda_{pre} \\ \lambda_{post} \end{pmatrix} \quad \text{(A2)}$$

Given the ease of computing (A2), and given that this requires a substantially lower computational burden to calculate than the weighted least squares regression, this is the formula which we use internally in `sdid.ado` to calculate $\widehat{\tau}^{sdid}$, all implemented directly in Mata. Thus, in practice the computational implementation of `sdid` (Pailañir and Clarke 2022) consists of firstly estimating weights $\omega$ and $\lambda$, and secondly calculating (A2).

## A4 References


Arkhangelsky, D., S. Athey, D. A. Hirshberg, G. W. Imbens, and S. Wager. 2021. Synthetic Difference-in-Differences. *American Economic Review* 111(12): 4088–4118. https://www.aeaweb.org/articles?id=10.1257/aer.20190159.

Pailañir, D., and D. Clarke. 2022. SDID: Stata module to perform synthetic difference-in-differences estimation, inference, and visualization. Statistical Software Components, Boston College Department of Economics. https://ideas.repec.org/c/boc/bocode/s459058.html.